VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi-590018

PROJECT REPORT ON

"AI BASED CAREER COUNSELLING"

Submitted in partial fulfillment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted By

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By,

Krisha M (1BY20CS089)

DECLARATION

I, **Krisha M [1BY20CS089]** student of VIII Semester BE, in the **Department of Computer Science and Engineering, BMS Institute of Technology and Management, Bengaluru** hereby declare that the Project work entitled "**AI BASED CAREER COUNSELLING"** has been carried out by us and submitted in partial fulfillment of the requirements for the VIII Semester degree of **Bachelor of Engineering in Computer Science and Engineering** of **Visvesvaraya Technological University**, Belagavi during academic year 2023-24.

ABSTRACT

This project introduces a groundbreaking approach to personalized career counseling through the development and deployment of an innovative chatbot system. Built upon contemporary programming languages and advanced analytical methodologies, the chatbot offers users bespoke guidance derived from personality assessments, aptitude evaluations, and career recommendations. In contrast to conventional counseling paradigms, this chatbot harnesses cutting-edge algorithms to deliver scalable, user-friendly, and data-centric insights into academic and professional trajectories .The paper commences with an exploration of the prevailing landscape of career counseling, spotlighting the myriad challenges individuals encounter while navigating intricate educational and vocational choices. A thorough literature review follows, encompassing extant studies on chatbot-driven counseling solutions and machine learning models for career prognostication. Ethical considerations and data security implications relevant to AI-powered counseling platforms are also deliberated. Detailed insights into the chatbot's architecture and functionalities are provided, elucidating technical intricacies such as data manipulation, model integration, and interface design. Implementation specifics, including code excerpts and deployment strategies, illuminate the developmental journey and scalability of the chatbot. Evaluation outcomes and results delineate user feedback, usability assessments, and performance metrics for personality profiling and career forecasting. Illustrative case studies underscore the chatbot's efficacy in influencing users' decision-making processes and the potency of personalized recommendations. The paper culminates in a discourse on the ramifications of AI-driven personalized career counseling, emphasizing strengths, constraints, and future research trajectories. Proposals for further refinement, algorithmic enhancements, and feature augmentation underscore the chatbot's potential to redefine the contours of career counseling.

Keywords: — Personalized career counselling, Chatbot system, Programming languages, Advanced analytical methodologies ,Personality assessments, Aptitude evaluations, Career recommendations, Machine learning models ,Data-centric insights ,Scalability ,User-friendly ,Data security ,Ethical considerations ,AI-powered counselling platforms, Interface design ,Usability assessments, Performance metrics, Personality profiling, Career forecasting, Decision-making processes.

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CHAPTER 1

INTRODUCTION

CHAPTER 1

INTRODUCTION

In a rapidly evolving world, the importance of making informed career choices has never been greater. The transition from secondary education to the broader landscape of higher learning and professional life is a pivotal moment in every student's journey. Yet, for many secondary-level students, the process of career decision-making can be daunting, complex, and often filled with uncertainties.

This project seeks to bridge this gap by introducing an innovative solution that leverages the power of Artificial Intelligence (AI) to provide comprehensive and accessible career counselling to students at the secondary level. The heart of this initiative is an AI-based career counselling chatbot, a virtual mentor capable of understanding, guiding, and inspiring students on their academic and vocational paths.

In traditional educational systems, career counselling services are often constrained by factors such as limited accessibility, high costs, and scalability challenges. The endeavour at hand aims to transform this landscape by introducing a technologically advanced and user-centric approach. This AI-driven chatbot will serve as an ever-present career counsellor in students' pockets, available 24/7 to offer timely and personalized guidance.

The project encompasses a multidimensional approach that involves data collection, AI model development, user interface design, and personalization algorithms. The chatbot will harness the latest advancements in Natural Language Processing (NLP) and machine learning, enabling it to understand and respond to user inquiries in a conversational manner. It will be complemented by an intuitive and accessible user interface, ensuring that students from diverse backgrounds, including those with disabilities, can comfortably access its services.

By introducing an AI-based career counselling chatbot to secondary-level students, this project endeavours to facilitate a transformative experience, equipping the next generation with the tools and knowledge to make educated career choices. It aspires to enhance accessibility, efficiency, and effectiveness in the realm of career guidance, fostering a brighter and more promising future for students as they embark on their academic and professional journeys.

1.1 Background

In India, the process of career decision-making among secondary school students has long been fraught with challenges, particularly concerning access to professional counselling services. Historically, career guidance in secondary education has heavily relied on traditional methods, with physical counselling sessions conducted by a limited number of counsellors in schools. However, the demand for such services far outweighs the available resources, especially in rural and underserved areas where access to quality education and career guidance is particularly limited. As a result, many students find themselves navigating the complexities of career choices without adequate support, leading to uncertainty and often ill-informed decisions regarding their future paths.

Moreover, the affordability of professional counselling services poses a significant barrier for many students and their families, exacerbating the inequities in accessing quality career guidance. Families from lower socio-economic backgrounds, in particular, may find it financially prohibitive to seek out private counselling services, further widening the gap in opportunities for informed decision-making. Consequently, students may resort to relying on generic recommendations or societal pressures in choosing their career paths, without fully considering their individual aptitudes, interests, and aspirations.

In recent years, there has been a growing recognition of the need to leverage technology to address the shortcomings of traditional career guidance methods. The advancements in artificial intelligence (AI) present a promising opportunity to democratize access to personalized career counselling services. By harnessing AI algorithms, such as Support Vector Machines (SVM), it becomes feasible to analyse students' aptitudes, personalities, and preferences at scale, providing tailored recommendations and career pathways remotely through digital platforms. This shift towards AI-based career counselling not only addresses the limitations of physical counselling services but also has the potential to reach a broader spectrum of students, including those in remote or marginalized communities, thereby fostering greater inclusivity and empowerment in career decision-making.

1.2 Literature Survey

1. Title: Artificial Intelligence Chat Bot for Counselling Therapy (Patil & Rasave, 2021)

   Description: Judges user's emotions through facial/text/voice analysis and provides confidence value on emotional state using TensorFlow

   Pros: Acts as simulation to help users express themselves; accurately maps emotional state

   Cons: Struggles to understand hidden feelings that can only be picked up in human interactions

2. Title: AI Based Career Counselling Chatbot (Zaidi et al., 2021)

   Description: Acts as career counsellor through anonymous chat to help students decide career paths using DialogFlow

   Pros: Reduces student stress; provides help at no cost 24/7; gives accurate answers

   Cons: Recommendations may not suit student interests and caliber

3. Title: Chatbot System for Mental Health Care (Das & Prasad)

   Description: Chatbot provides conversations and motivation for better mental health using CBT therapy and mood tracking via NLP

   Pros: Helps people track moods; clinically tested with success

   Cons: Should not fully replace human therapists for severe cases

4. Title: Towards a Chatbot for Digital Counselling (Cameron et al., 2017)

   Description: Proposes emoji-based chatbot for mental health counselling to assess user issues and provide tips

   Pros: Transforms digital counselling into more empathetic experience

   Cons: Scope for improvement in user-friendly design

5. Title: Artificial Intelligence for Career Guidance – Current Requirements and Prospects for the Future (Westman et al., 2021)

Description: Explores the use of AI in higher education career guidance through focus groups, scenarios, and trials. AI acted as information providers and analytics tools to support students and staff.

Pros: Enables personalized and proactive interventions; frees up staff time from routine tasks; provides career insights through data analysis

Cons: Lacks holistic learner profiles; struggles with relevance and accuracy of suggestions; raises ethical concerns around data privacy and bias

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   Cons: Lacks holistic learner profiles; struggles with relevance and accuracy of suggestions; raises ethical concerns around data privacy and bias

7. "Artificial Intelligence for Career Guidance – Current Requirements and Prospects for the Future" by Barnes, et al. (2020) in the ERIC Journal

   This paper explores the potential of artificial intelligence (AI) to enhance career guidance services and discusses the current requirements and challenges of implementing AI-based career guidance systems.

8. "Scrutinising Artificial Intelligence based Career Guidance and Counselling Systems: an Appraisal" by Singh, et al. (2022) in the International Journal of Engineering and Technology

   This paper reviews the current state of AI-based career guidance and counselling systems and identifies the key challenges and opportunities for future development. The authors identify several potential benefits of using AI in career guidance, such as the ability to provide personalized recommendations, the ability to access a wider range of information and resources, and the ability to automate many of the tasks that are currently performed by human career counsellors.

9. Title: "Navigating Futures: Understanding Career Decision-Making Self-Efficacy in Grade 11 Students"

   Pros: Comprehensively integrating parental attachment, dysfunctional career thoughts, and career exploration, this study offers valuable insights into the multifaceted determinants of career decision-making self-efficacy.

   Cons: The limited generalizability to Grade 11 students restricts broader applicability, and methodological constraints, including reliance on self-report measures and a correlational design, pose challenges in establishing causation; furthermore, the cross-sectional nature lacks a temporal dimension, hindering a nuanced understanding of the dynamic aspects of career decision-making self-efficacy.

1.3 Motivation

Every individual in India faces the daunting task of choosing the right career path after completing their 10th standard education. We understand this challenge firsthand, having been in the same shoes ourselves. Recognizing the widespread nature of this issue, we are motivated to embark on this project to alleviate the struggles faced by students across the country.

The decision-making process post-10th grade is pivotal, setting the trajectory for one's academic and professional journey. However, the lack of accessible and comprehensive career guidance often leaves students feeling overwhelmed and uncertain about their future prospects. We believe that every student deserves the opportunity to make informed decisions about their career path, regardless of their background or circumstances.

By developing an AI-driven career counseling chatbot, we aim to democratize access to personalized guidance and support for students at the secondary level. Our motivation stems from a deep-seated desire to empower individuals with the tools and knowledge they need to navigate the complexities of the modern job market successfully.

Through this project, we aspire to make a meaningful difference in the lives of students across India, ensuring that they are equipped to make educated and confident choices about their academic and vocational paths. We believe that by addressing this critical need, we can contribute to the creation of a more empowered and prosperous future for all individuals in India.

1.4 Problem Statement

In the realm of secondary education, the process of making career choices is often marred by limited access to professional counselling services, particularly in rural or underserved areas, leading to generic recommendations that fail to consider individual strengths, weaknesses, and evolving interests. Additionally, there is a notable absence of widespread integration of AI-driven solutions for career guidance, despite significant advancements in technology. Consequently, students face a lack of confidence in their career choices, highlighting an urgent need for accessible AI-based counselling, aptitude tests, and detailed career paths to empower every secondary school student with tailored guidance for navigating their future paths effectively.

1.5 Scope

The scope of this project in the context of India addresses the prevalent challenge of limited access to affordable career counselling services, compounded by a scarcity of counsellors. Leveraging AI-based career counselling becomes imperative to bridge this gap effectively. By employing AI technologies, specifically utilizing Support Vector Machine (SVM) algorithms, we aim to analyse students' aptitudes, characteristics, and personality types comprehensively. Through a dedicated career chatbot, personalized career recommendations will be provided, considering individual strengths, weaknesses, and evolving interests. The scope encompasses the development of an accessible platform or application that enables students across diverse socio-economic backgrounds to access career guidance remotely. Additionally, the project will involve partnerships with educational institutions and government agencies to ensure widespread adoption and sustainability. Through this initiative, we aim to empower students with tailored career paths, thus fostering confidence and informed decision-making regarding their future endeavours.

1.6 Aim and Objectives

Aim of the Project:

1 Enhance Accessibility: The primary aim of this project is to enhance accessibility to quality career guidance for secondary school students across India, particularly those in rural and underserved areas, where physical counselling services are limited or inaccessible.

2 Empower Students: Empower students to make well-informed career decisions by providing personalized, AI-driven counselling services that take into account individual aptitudes, interests, and personality traits.

3 Utilize AI Technology: Leverage advancements in artificial intelligence (AI) technology, including Support Vector Machine (SVM) algorithms, to analyse student data and provide accurate career recommendations and pathways.

4 Bridge the Gap: Bridge the gap between the demand for career counselling services and the limited availability of professional counsellors by providing a scalable and sustainable digital platform for career guidance.

5 Promote Inclusivity: Promote inclusivity by ensuring that the project is accessible to students from diverse socio-economic backgrounds, linguistic communities, and geographic locations, thereby mitigating disparities in access to career guidance.

6 Foster Confidence: Foster confidence among students in their career choices by offering personalized support and guidance tailored to their unique strengths, interests, and aspirations.

7 Facilitate Career Planning: Facilitate comprehensive career planning by providing students with clear and actionable pathways towards their chosen career goals, including information on educational requirements, skill development opportunities, and employment prospects.

Objectives of the Project:

1 Develop AI-driven Platform: Develop a user-friendly digital platform equipped with AI-driven counselling capabilities, including chatbots and recommendation systems, to deliver personalized career guidance services to secondary school students.

2 Conduct Needs Assessment: Conduct a comprehensive needs assessment to identify the specific challenges and requirements of students regarding career guidance, taking into account factors such as geographical location, socio-economic status, and educational background.

3 Train AI Algorithms: Train AI algorithms, such as SVM, using robust datasets comprising student profiles, aptitude test results, career pathways, and other relevant variables, to ensure the accuracy and reliability of career recommendations.

4 Design Interactive Content: Design interactive content, including aptitude tests, career quizzes, informational resources, and multimedia presentations, to engage students and facilitate self-discovery and exploration of career options.

5 Establish Partnerships: Establish partnerships with educational institutions, government agencies, NGOs, and corporate entities to promote the adoption and dissemination of the AI-driven career counselling platform and ensure its sustainability.

6 Pilot Testing and Evaluation: Conduct pilot testing of the platform in select schools and communities to gather feedback, assess usability, and measure the effectiveness of the counselling services in influencing students' career decisions and aspirations.

7 Scale Up and Expansion: Scale up the project to reach a wider audience of secondary school students across India, expanding the reach and impact of the AI-driven career counselling platform through strategic partnerships, outreach initiatives, and continuous improvement based on feedback and evaluation results.

1.7 Challenges

The limitations of existing career counselling systems, especially those not utilizing advanced AI-driven chatbots include:

1. Limited Accessibility: Traditional career counselling services are often limited in terms of availability, making them inaccessible to students in remote or underserved areas.
2. High Costs: In-person career counselling can be expensive, and students from low-income backgrounds may struggle to afford such services.
3. Scalability: Traditional counselling services may struggle to scale their operations to meet the needs of a large number of students.
4. Generic Advice: Many existing systems provide one-size-fits-all advice, which may not consider an individual's unique strengths, weaknesses, and interests.
5. Lack of Personalization: Personalization is often minimal in traditional systems, making it challenging to provide tailored career guidance.
6. Time Constraints: In-person counselling is often time-constrained, limiting the depth of discussions and interactions with students.
7. Resource Dependency: Traditional systems may heavily rely on human counsellors, leading to resource constraints and the inability to serve all students effectively.
8. Limited Data Analysis: Traditional systems may not make effective use of data analysis for trend identification and career recommendation improvements.
9. Bias and Subjectivity: Human counsellors can introduce bias and subjectivity into the counselling process, potentially affecting the quality and fairness of advice.

CHAPTER 2

OVERVIEW

CHAPTER 2

OVERVIEW

The project aims to revolutionize career guidance and counseling for secondary school students in India by leveraging AI-driven solutions. The current scenario in India highlights significant challenges in accessing professional counseling services, particularly in rural and underserved areas. Furthermore, the limited availability of counselors exacerbates the issue, leaving many students ill-prepared to make informed career decisions. To address these challenges, the project introduces an innovative AI-based career counseling platform that provides personalized guidance tailored to individual aptitudes, interests, and personality traits.

Key Components of the Project:

1. AI-driven Counseling Platform: The cornerstone of the project is the development of an AI-driven counseling platform that utilizes advanced algorithms, including Support Vector Machine (SVM), to analyze student data and provide personalized career recommendations. This platform serves as a digital alternative to traditional counseling services, offering accessibility to students regardless of their geographical location or socio-economic background.
2. User Interaction: The platform engages students through interactive sessions where they respond to a series of questions related to their preferences, aptitudes, and personality traits. These questions cover a range of topics, including social interactions, decision-making, and problem-solving skills, allowing the AI system to generate a holistic profile of each student.
3. Career Recommendation: Based on the information gathered from the user responses, the AI system predicts suitable career paths for the students. This prediction takes into account various factors, including academic interests, career aspirations, and market demand, to provide accurate and relevant recommendations.
4. OCEAN Personality Assessment: In addition to career guidance, the platform also assesses students' personality traits using the OCEAN model (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism). By understanding their personality profiles, students gain valuable insights into their strengths and weaknesses, which further inform their career decisions.

Overall, the project represents a paradigm shift in the way career counseling is delivered to secondary school students in India, offering a comprehensive and accessible solution that empowers students to make informed decisions about their future paths. By harnessing the power of AI technology, the project seeks to democratize career guidance and bridge the gap between students' aspirations and the opportunities available to them.

CHAPTER 3

SYSTEM REQUIREMENT SPECIFICATION

CHAPTER 3

SYSTEM REQUIRIMENT SPECIFICATION

3.1 Functional Requirements

1. Design and implementation of user interface for student interaction – We provide a user-friendly interface which is definitely appealing for the students and interests them to engage in the interface.
2. Creation of MBTI assessment module with 20 questions - The Myers–Briggs Type Indicator is a pseudoscientific self-report questionnaire that claims to indicate differing "psychological types". The test attempts to assign a binary value to each of four categories: introversion or extraversion, sensing or intuition, thinking or feeling, and judging or perceiving. Development of SVM model for MBTI classification – Support Vector Machine is one of the supervised learning models that is used for MBTI classification to give accurate predictions and results.
3. Implementation of OCEAN personality assessment with 50 questions- These five primary personality traits are known as the acronym OCEAN, for openness, conscientiousness, extraversion, agreeableness and neuroticism. The OCEAN model supports that, regardless of gender, age or nationality, all of us are made up of 5 basic traits.
4. Visualization of OCEAN personality traits using bar plot representation – Visualization is one of the main feature to analyze the output, therefore with the help of bar plot we are able to analyze the personality traits of the student.
5. Design and inclusion of aptitude test with 10 questions – Aptitude tests are used to check the technicalities of understanding. We need to invest time to solve them therefore they check our strengths, of how well the concept is being understood.
6. Calculation of total aptitude score based on student responses – The total aggregate is calculated and the final aptitude score is being calculated.
7. Development of career recommendation algorithm based on personality type – Now after getting a overall idea about the students based on the tests, an algorithm recommends a career that suits the personality.
8. Integration of career database for providing recommended career options – Out of all the possible options available that is stored in the database, the appropriate one is selected and presented to the student.

9. Implementation of feedback mechanism for user input on recommendations -It is very much necessary to collect the users feedback for any further improvements.
10. Optimization of user experience for seamless interaction and navigation – We must make sure the user had a very smooth experience using the website and therefore the components are made with utmost importance.

3.2 Non-Functional Requirements

1. Performance: The system should respond to user queries promptly, with minimal latency, to ensure a smooth user experience.
2. Scalability: The application should be able to handle increasing numbers of concurrent users without significant degradation in performance.
3. Reliability: The system should be robust and dependable, with minimal downtime or service interruptions.
4. Compatibility: The application should be compatible with a wide range of devices and web browsers to ensure accessibility for all users.
5. Maintainability: The codebase should be well-structured and documented, allowing for easy maintenance, updates, and bug fixes.
6. Usability: The user interface should be intuitive and user-friendly, with clear instructions and navigation paths to facilitate ease of use for students.

3.3 User Requirements

1. Integration of OpenAI GPT-3.5 for natural language processing.
2. Design and implementation of user interface for student interaction.
3. Creation of MBTI assessment module with 20 questions.
4. Development of SVM model for MBTI classification.
5. Implementation of OCEAN personality assessment with 50 questions.
6. Visualization of OCEAN personality traits using bar plot representation.
7. Design and inclusion of aptitude test with 10 questions.
8. Calculation of total aptitude score based on student responses.
9. Development of career recommendation algorithm based on personality type.
10. Integration of career database for providing recommended career options.
11. Implementation of feedback mechanism for user input on recommendations.
12. Optimization of user experience for seamless interaction and navigation.

3.4 Domain Requirements

1. Understanding of career development and guidance principles.
2. Knowledge of personality assessment theories (e.g., MBTI, OCEAN).
3. Proficiency in natural language processing (NLP) techniques.
4. Familiarity with machine learning algorithms for classification (e.g., SVM).
5. Access to a comprehensive career database for recommended career options.
6. Awareness of user experience design principles for effective interaction design.
7. Understanding of web development technologies for user interface implementation.
8. Ability to visualize data using appropriate graphical representations (e.g., bar plots).
9. Knowledge of software engineering practices for maintainable code development.
10. Understanding of performance optimization techniques for responsive system behavior.
11. Familiarity with feedback mechanisms for user input collection and analysis.

3.5 System Requirements

1. Adequate computational resources to support the integration and execution of OpenAI GPT-3.5 for natural language processing.
2. Development environment for web application implementation (e.g., PyCharm for Flask application).
3. Access to relevant libraries and frameworks for machine learning development (e.g., scikit-learn for SVM model development).
4. Storage capacity for storing user data, assessment responses, and career database.
5. Compatibility with web browsers to ensure accessibility for users across different platforms (e.g., Chrome, Firefox, Safari).
6. Web server for hosting and serving the Flask application.
7. Network connectivity for accessing external resources such as OpenAI API and career databases.

CHAPTER 4

DETAILED DESIGN

CHAPTER 4

DETAILED DESIGN

4.1 Overall Design

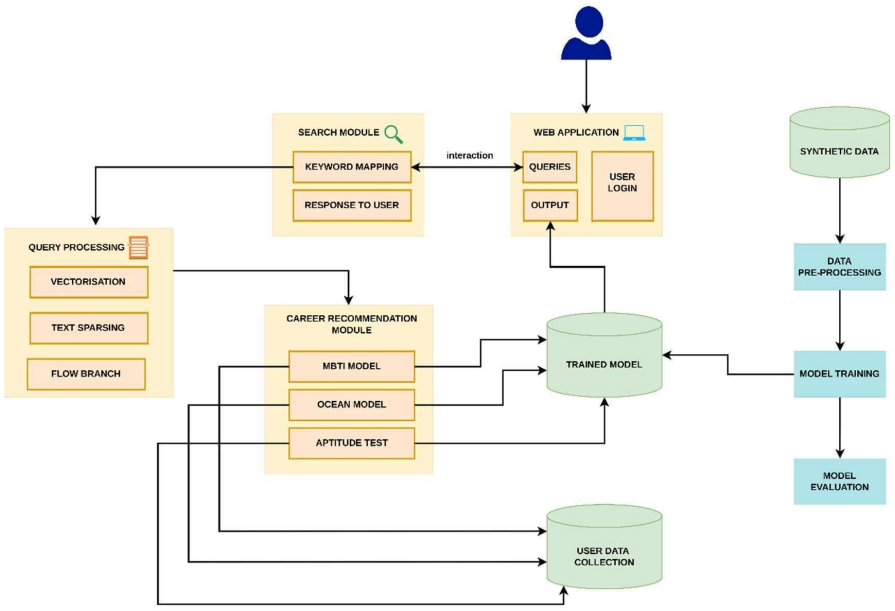


Fig 4.1.: Overall design

1. User Input: The process begins with a user entering a query into the web application. This querycould be anything related to careers, such as keywords or descriptions.

2. Search Module: The web application then sends the user's query to a search engine, which is likelynot shown in detail here but is represented by the block labeled "Search Module."

3. Data Processing: The search engine returns results, which may include data about MBTI (Myers- Briggs Type Indicator) and OCEAN personality types.

4. AI model and Career Recommendation: Once processed, the data is fed into a model, possibly an AI model, for careerrecommendation. The model considers the user's MBTI or OCEAN types to generate suitable careerrecommendations.

5. User Output: The web application displays these career recommendations to the user.

6. Data Store: These data stores contain information about MBTI responses, OCEAN responses, a training model, MBTI dataset.

4.2 Activity Diagram

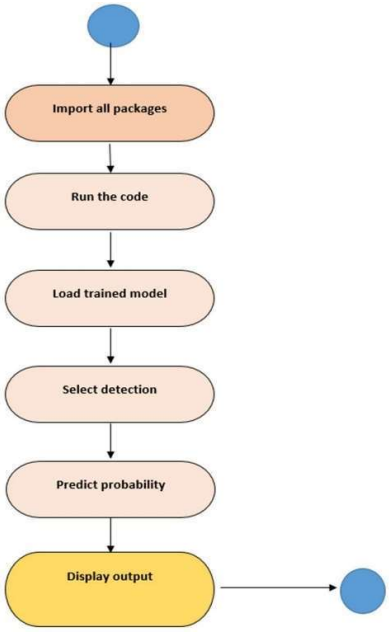


Fig 4.2.: Activity diagram

An activity diagram is a behavioral diagram i.e. it depicts the behavior of a system. An activity diagram portrays the control flow from a start point to a finish point showing the various decision paths that exist while the activity is being executed.

Firstly, we import all the library package necessary to run the code and for supporting the code to run error free. As soon as code is run it provides the desired output.

4.3 Use Case Diagram

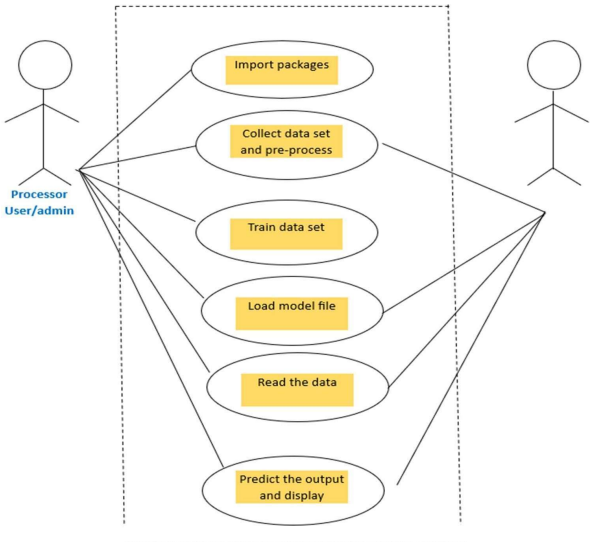


Fig 4.3.: Use case diagram

Use case consists of user and processor where user is used to provide the input to the system and

processor is used to process the input data and provide output. The flow is shown in the above diagram. First user as to run the system and run the code, model and library packages are imported and loaded. After the run of code output is displayed according to the data input provided.

4.4 Class Diagram

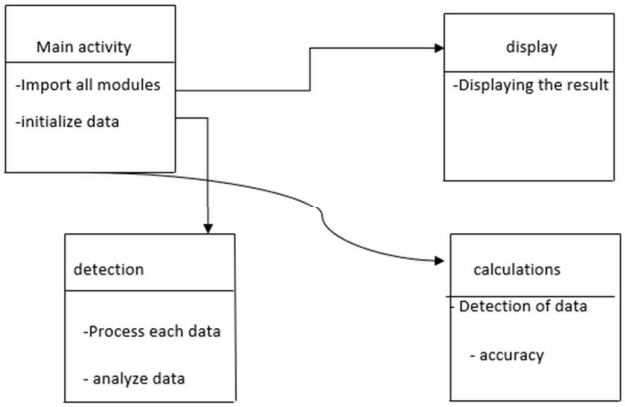


Fig 4.4.: Class diagram

The class diagram depicts a static view of an application. It represents the types of objects residing in the system and the relationships between them. A class consists of its objects, and also it may inherit from other classes. A class diagram is used to visualize, describe, document various different aspects of the system, and also construct executable software code.

It shows the attributes, classes, functions, and relationships to give an overview of the software system. It constitutes class names, attributes, and functions in a separate compartment that helps in software development. Since it is a collection of classes,interfaces, associations, collaborations, and constraints, it is termed as a structural diagram.

4.5 Sequence Diagram

Sequence diagram consists of 5 different blocks namely user, processor, scaling, Model and labels as shown in the below figure.

User will provide the input data through the image files already saved in the system wherepreprocessing of data is done which is differentiator parameters and after that those are stored in the memory unit.

After preprocessing and storing of dataset is done, trained model file is loaded where the featured of the file is extracted for classifying the output. After classifying the output, Prediction values are displayed.

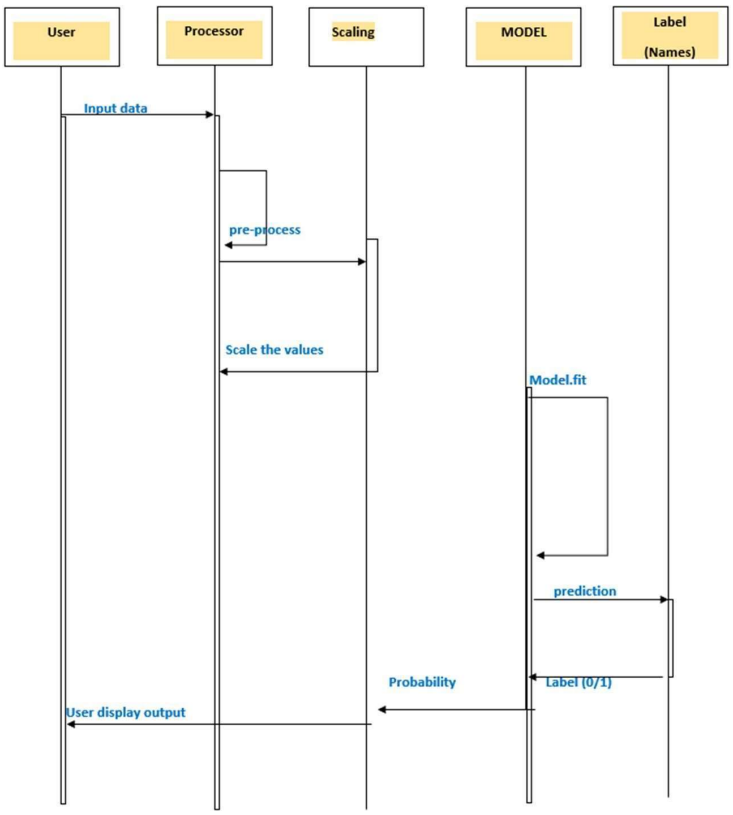


Fig 4.5.: Sequence Diagram

4.6 Data Flow Diagram Level 0:

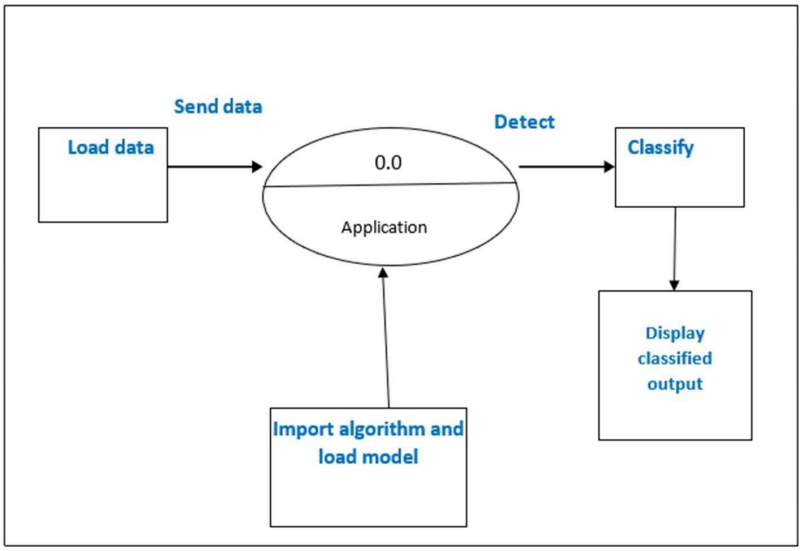


Fig 4.6.: DFD 0

Above mentioned diagram is the representation of DFD0 which provides u the content diagram or say overview of the whole system. It is designed to be an at- a-glance view, showing the system as single

high-level process. Here from the file data is be loaded tothe application where the loaded data is sent to classification unit to predict the result withthe help of trained model file and output is known.

4.7 Data Flow Diagram Level 1:

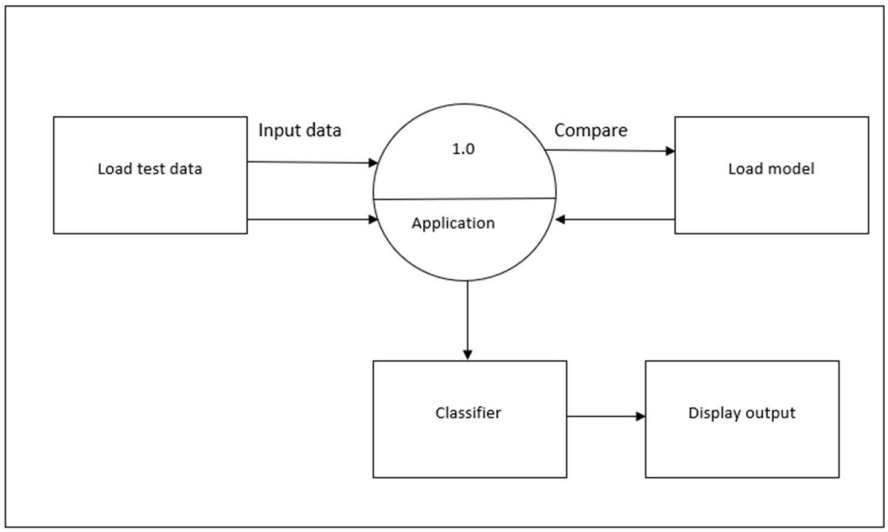


Fig 4.7.: DFD 1

Above mentioned diagram is the representation of DFD1. The Level 0 DFD is broken down into more specific, Level 1 DFD. Level 1 DFD depicts basic modules in the system and flow of data among various modules. Here from the file data is be loaded to the application where the loaded data is sent to classification unit to predict the result and classes are classified given a label.

CHAPTER 5

IMPLEMENTATION

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IMPLEMENTATION

5.1 Programming Language



Fig 5.1.: Python Programming Language

The project utilizes Python is the main language for programming because of capabilities, popularity, extensive libraries for machine learning and web development, and ease of integration with deep learning frameworks. Python provides a robust ecosystem for data manipulation, image processing, model development, and web application development, making it well-suited for the prerequisites of the project. Python is chosen as the primary programming language for this project due to a multitude of reasons, each catering to different aspects of the project's requirements:

1. **Popularity**: Python enjoys widespread popularity in data science, machine learning, and web development. It is popular because of its simplicity, readability, and versatility. A vast community of developers actively contributes to its ecosystem by creating libraries, frameworks, and tools.
2. **Extensive Libraries for Machine Learning:** Python boasts an extensive collection of libraries especially designed for activities involving machine learning tasks. Libraries such as TensorFlow, Keras, PyTorch, and Scikit-learn provide robust tools and frameworks for developing, training, building implementing models for machine learning with ease. These libraries offer pre-implemented algorithms, customizable modules, and seamless using other data sources processing libraries, streamlining the development process.
3. **Rich Ecosystem for Web Development:** Python's web development ecosystem, primarily driven by frameworks like Django and Flask, offers comprehensive solutions for building web applications. These frameworks provide built-in functionalities for handling routing, templating,

database management, and user authentication, enabling developers to work on implementing application logic as opposed to handling low-level infrastructure tasks.

4. **Ease of Integration accompanied by Deep Learning Frameworks**: Python easily integrates with well- known deep learning frameworks, facilitating the development of sophisticated neural network architectures for tasks such as picture classification, object detection, and natural language processing. Frameworks like TensorFlow and PyTorch provide high-level APIs and abstractions, enabling developers to prototype and experiment with complex neural network architectures efficiently.
5. **Robust Ecosystem backing Data Manipulation and Image Processing:** Python offers powerful libraries such as NumPy, Pandas, and OpenCV for data manipulation, analysis, and image processing. These libraries provide efficient data structures, array manipulation functions, and image processing algorithms, empowering developers to preprocess and augment datasets effectively before feeding them into machine learning models.

5.2 Algorithms

i. Support Vector Machines (SVM):

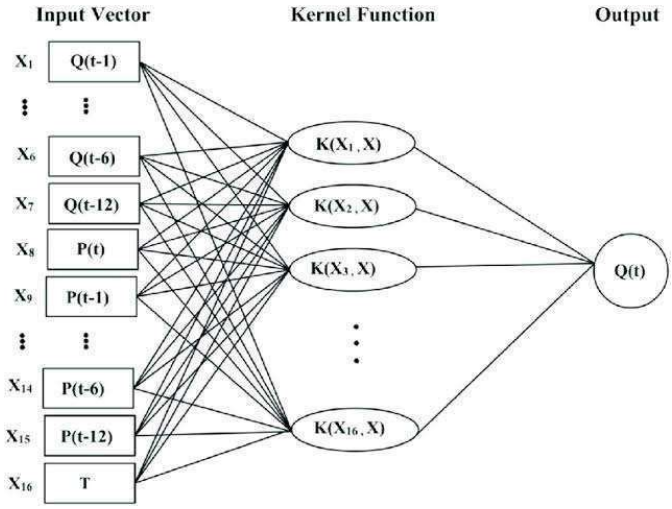


Fig 5.2.: SVM Architecture

Support Vector Machines (SVMs) are adept at career recommendation tasks, leveraging individuals' diverse attributes like skills, education, experience, and preferences to suggest suitable career paths. By analyzing high-dimensional data, SVMs determine optimal decision boundaries between various career categories, providing personalized recommendations tailored to individuals' unique profiles and aspirations. This enables students, job seekers, or those considering career transitions to make

well-informed decisions about their professional trajectories.

In career recommendation systems, Support Vector Machines (SVMs) serve as robust algorithms capable of effectively analyzing multifaceted individual profiles to offer tailored career guidance. By processing factors such as educational background, skill sets, work experience, and personal preferences, SVMs discern intricate patterns in the data to classify individuals into suitable career categories. Their ability to handle high-dimensional data and nonlinear relationships enables SVMs to deliver accurate and personalized recommendations, empowering individuals to navigate their career paths with confidence and clarity.

ii. Large Language Models (LLM):

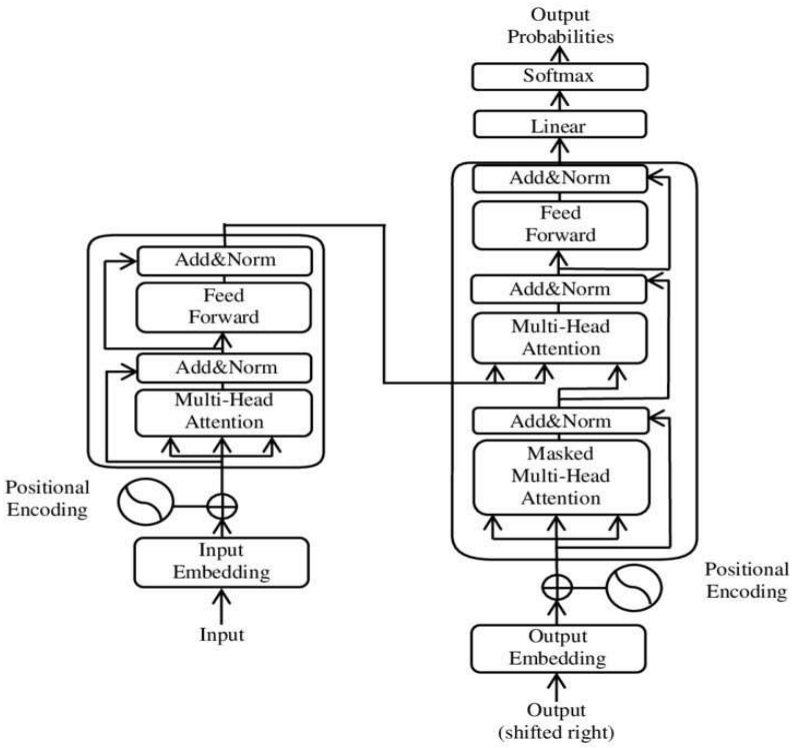


Fig 5.3.: – LLM Model Architecture

Large Language Models (LLMs), such as GPT (Generative Pre-trained Transformer) models, play a pivotal role in career recommendation systems by processing diverse sets of individual data to offer personalized career guidance. By analyzing factors like educational background, work experience, skills, personal interests, and career aspirations, LLMs generate tailored recommendations suited to an individual's unique profile. Leveraging vast amounts of textual data, LLMs can discern nuanced patterns and trends in career trajectories, enabling them to provide insightful suggestions for career

exploration and advancement. Through their adaptability and deep understanding of language semantics, LLMs empower individuals to make informed decisions about their professional paths, facilitating a more personalized and effective approach to career planning and development.

Large Language Models (LLMs), like GPT models, are instrumental in career recommendation systems, analyzing diverse individual data to offer personalized guidance. By processing factors such as education, skills, experience, and aspirations, LLMs generate tailored career suggestions. Their deep understanding of language semantics enables nuanced recommendations, empowering individuals to navigate their professional journeys effectively.

5.3 Concepts Used

i. OCEAN:

Incorporating OCEAN (Big Five personality traits) analysis enriches the Chatbot's capabilities by providing deeper insights into users' personality characteristics. Similar to MBTI integration, this process begins with acquiring a dataset containing individuals' scores on the Big Five traits. Preprocessing the dataset involves extracting relevant features and formatting the data for analysis or model training. Integration may involve employing machine learning techniques or statistical analysis to assess users' personality traits based on their input or interactions with the Chatbot. By understanding users' OCEAN profiles, the Chatbot can offer more personalized career guidance aligned with their individual preferences and tendencies. Ethical considerations, such as data security and ensuring user consent for personality analysis, remain paramount throughout the integration process to uphold user trust and privacy.

ii. MBTI :

For integrating MBTI functionality, the first step is to procure a dataset containing individuals' MBTI types. This dataset serves as the foundation for training machine learning models, such as Support Vector Machines (SVM), to predict MBTI personality types based on input data. Preprocessing the dataset involves feature extraction and formatting to prepare it for model training. Once trained, the SVM model can be seamlessly integrated into the application's logic, enabling the Chatbot to provide personalized career recommendations tailored to users' MBTI profiles. Ethical considerations, including transparency about data usage and user consent, are essential throughout this process to ensure the responsible handling of sensitive user information.

iii. GPT-3.5 :

With GPT-3.5, the focus is on preparing text prompts and managing responses generated by the model. It's essential to handle these interactions ethically, considering factors such as data privacy and responsible use of AI technology. By seamlessly integrating GPT- 3.5, the Career Chatbot can engage users in natural language conversations, enhancing their experience and providing valuable guidance tailored to their needs and queries.

5.4 Proposed Design

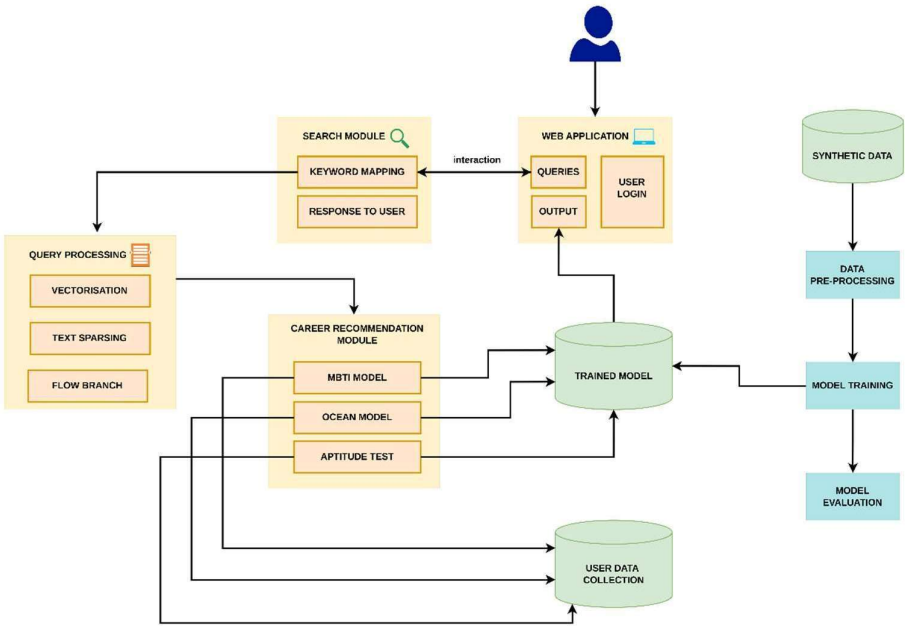


Fig 5.4.: System Architecture

1. User Input: The process begins with a user entering a query into the web application. This querycould be anything related to careers, such as keywords or descriptions.

2. Search Module: The web application then sends the user's query to a search engine, which is likelynot shown in detail here but is represented by the block labeled "Search Module."

3. Data Processing: The search engine returns results, which may include data about MBTI (Myers- Briggs Type Indicator) and OCEAN personality types.

4. AI model and Career Recommendation: Once processed, the data is fed into a model,

possibly an AI model, for careerrecommendation. The model considers the user's MBTI or OCEAN types to generate suitable careerrecommendations.

5. User Output: The web application displays these career recommendations to the user.

6. Data Store: These data stores contain information about MBTI responses, OCEAN responses, a training model, MBTI dataset.

5.5 Code

5.5.1 SVM Model Training:

```
import joblib     #for loading and saving trained models

import pandas as pd       #for creating and handling dataframes

import seaborn as sns     #for advanced plots

import matplotlib.pyplot as plt   #for plotting
#DATA COLLECTION

data = pd.read_csv('synthetic_data.csv')


# print("shape of the given dataset is : ",data.shape)#gives shape of dataset  #

print("size of the given dataset is : ",data.size)#gives size of dataset

# print(data['mbti_type'].unique())



#FEATURE SELECTION

x = data.iloc[:,:-1].values#choosing input columns y =

data.iloc[:,-1].values#choosing output column
```

```
#FEATURE ENGINEERING

from sklearn.model_selection import train_test_split

#splitting dataset into train and test partitions for evaluating model after training

x_train,x_test,y_train,y_test = train_test_split(x,y,random_state=100,test_size=0.3)


from sklearn.preprocessing import StandardScaler sc =

StandardScaler()

#scaling the data

x_train_scaled = sc.fit_transform(x_train) x_test_scaled =

sc.transform(x_test)

joblib.dump(sc, '1_scaler.pkl')#saving the scaler into hard disk


#MODEL TRAINING
from sklearn.svm import SVC #importing algorithm

svm_model = SVC(probability=True)    #initialising the algorithm

svm_model.fit(x_train_scaled,y_train)    #training the algorithm on dataset joblib.dump(svm_model,

'1_svm_model.pkl')      #saving the trained model


#MODEL EVALUATION

from sklearn.metrics import classification_report

y_pred = svm_model.predict(x_test_scaled)      #creating a classification report

print("classification report : \n",classification_report(y_test,y_pred))      #displaying the
classification report


from sklearn.metrics import confusion_matrix
```

```
svm_cf = confusion_matrix(y_test,y_pred)        #creating a confusion matrix


classes = ['INTP', 'ENTP', 'ISFP', 'INFJ',

'ESTJ', 'ENFP', 'ESTP', 'INFP',

'ISTP', 'ENTJ', 'ESFJ', 'ESFP',

'ISFJ', 'INTJ', 'ENFJ', 'ISTJ']

sns.heatmap(svm_cf,annot=True,fmt='d',cmap='Blues',        xticklabels=classes,yticklabels=classes)

          #plotting the confusion matrix as a heatmap

plt.savefig('svm_confusion_matrix.png',bbox_inches='tight')       #saving the confusion matrix

plt.title("SVM CONFUSION MATRIX")

plt.show()#displaying the plot
```

5.5.2 Application:

```
import joblib

import   matplotlib.pyplot   as   plt   import

matplotlib

import seaborn as sns

from flask import Flask, render_template, request, redirect, send_file from flask

import url_for

import os import io

from openai import OpenAI

openai_client                                =                                OpenAI(api_key="sk-
RYlIWBiGDpHDpC6lDuQCT3BlbkFJ42yrKcw81JOjPSuAOTUE")

matplotlib.use('Agg') app = Flask(

name__) initial_questions = [

"mbti20. I prefer small gatherings over large parties.", "mbti19. I feel
```

energized after spending time alone.",

"mbti18. I enjoy one-on-one conversations more than group discussions.", "mbti17. I need time to recharge after socializing.",

"mbti16. I am comfortable with long periods of solitude.", "mbti15. I do not often trust my instincts and gut feelings.",

"mbti14. I do not enjoy brainstorming and coming up with new ideas.", "mbti13. I am not drawn to abstract and theoretical concepts.", "mbti12. I find routine and predictability comforting.",

"mbti11. I rely on my practical experiences and observations.", "mbti10. I prioritize logic and reason in decision-making.",

"mbti9. I find it easy to detach myself emotionally from situations.",
"mbti8. I value honesty and directness over tactfulness.", "mbti7. I prefer to make decisions based on objective criteria.", "mbti6. I am more analytical than empathetic.",

"mbti5. I like to plan and organize my activities in advance.",

"mbti4. I feel uncomfortable with uncertainty and prefer a clear plan.", "mbti3. I enjoy setting and achieving specific goals.",

"mbti2. I prefer to have a routine and stick to it.",

"mbti1. I find it satisfying to complete tasks before deadlines."

]

ocean = [

"ocean50. I enjoy trying new and unfamiliar activities.", "ocean49. I am curious about different cultures and ways of life.", "ocean48. I appreciate art and creativity.",

"ocean47. I often come up with novel ideas and solutions.", "ocean46. I enjoy exploring abstract concepts and theories.", "ocean45. I am open to unconventional beliefs and ideas.", "ocean44. I enjoy discussing philosophical and theoretical topics.", "ocean43. I am interested in trying new and

exotic foods.", "ocean42. I enjoy reading fiction and exploring imaginary worlds.",
"ocean41. I am open to changing my opinions based on new information.", "ocean40. I am organized and keep my belongings in order.",
"ocean39. I set clear goals for myself and work towards them.", "ocean38. I am disciplined and stick to my plans.",
"ocean37. I pay attention to detail and strive for accuracy.", "ocean36. I prioritize my tasks and manage my time effectively.",
"ocean35. I am reliable and fulfill my commitments.", "ocean34. I find it easy to stay focused on my work.", "ocean33. I prefer to plan ahead rather than be spontaneous.", "ocean32. I am careful and cautious in my decision-making.", "ocean31. I take my responsibilities seriously.",
"ocean30. I enjoy socializing and being around people.", "ocean29. I am outgoing and enjoy meeting new people.", "ocean28. I feel comfortable in group settings and parties.", "ocean27. I often take the initiative in social situations.", "ocean26. I enjoy being the center of attention at times.", "ocean25. I am talkative and enjoy expressing my thoughts.",
"ocean24. I prefer to spend time with others rather than being alone.", "ocean23. I find social events and gatherings energizing.",
"ocean22. I enjoy participating in group activities and team projects.", "ocean21. I often seek out social engagements and opportunities.", "ocean20. I am considerate of others' feelings.",
"ocean19. I enjoy helping others and providing support.", "ocean18. I value cooperation and collaboration.",
"ocean17. I avoid conflicts and disagreements when possible.", "ocean16. I am empathetic and understanding towards others.", "ocean15. I am forgiving and don't hold grudges easily.", "ocean14. I try to be polite and considerate in my interactions.", "ocean13. I find it easy to trust others.",
"ocean12. I enjoy working in collaborative and team-oriented environments.", "ocean11. I value

harmony and avoid confrontation.",

"ocean10. I am prone to worry and anxiety.",

"ocean9. I often feel stressed about various aspects of my life.", "ocean8. I

am easily upset or irritated.",

"ocean7. I experience mood swings and emotional highs/lows.", "ocean6. I

tend to dwell on negative experiences.",

"ocean5. I often feel anxious about the future.", "ocean4. I am easily

stressed in challenging situations.",

"ocean3. I am sensitive to criticism and negative feedback.", "ocean2. I find

it difficult to let go of worries.",

"ocean1. I experience a wide range of emotions in my daily life.",

"apti10. If a car travels at a speed of 60 km/hour, how many kilometers will it travel in 5 hours?",

"apti9. A shopkeeper sells a shirt for $25, making a profit of 25%. What is the cost price of the shirt?",

"apti8. If 3 pencils cost $0.75, how much will 8 pencils cost?",

"apti7. If the perimeter of a square is 36 cm, what is the length of one side?",

"apti6. If 4 workers can build a wall in 4 days, how many days will it take for 8 workers to build the same wall?",

"apti5. A train travels at a speed of 80 km/hour. How long will it take to travel 320 kilometers?",

"apti4. If the area of a rectangle is 60 square meters and its length is 12 meters, what is its width?",

"apti3. If the selling price of an item is $120 and the profit earned is 20%, what is the cost price of the item?",

"apti2. If 1/4 of a number is 15, what is the number?",

"apti1. If a box contains 12 chocolates and 3 chocolates are taken out, what percentage of chocolates remains in the box?"]

aptitude["apti10. If a car travels at a speed of 60 km/hour, how many kilometers will it travel in 5 hours?",

```
"apti9. A shopkeeper sells a shirt for $25, making a profit of 25%. What is the cost price of the shirt?",

"apti8. If 3 pencils cost $0.75, how much will 8 pencils cost?",

"apti7. If the perimeter of a square is 36 cm, what is the length of one side?",

"apti6. If 4 workers can build a wall in 4 days, how many days will it take for 8 workers to build the same wall?",

"apti5. A train travels at a speed of 80 km/hour. How long will it take to travel 320 kilometers?",

"apti4. If the area of a rectangle is 60 square meters and its length is 12 meters, what is its width?",

"apti3. If the selling price of an item is $120 and the profit earned is 20%, what is the cost price of the item?",

"apti2. If 1/4 of a number is 15, what is the number?",

"apti1. If a box contains 12 chocolates and 3 chocolates are taken out, what percentage
of chocolates remains in the box?"

]

aptitude_answers = [300, 20, 2, 9, 2,

4, 5, 100, 60, 75]

# Global variable to track whether in recommendation mode

recommendation_mode = False

# Global lists to store user responses

user_responses = [] ocean_responses = []

aptitude_responses = []



# Counter to keep track of the OCEAN questions

ocean_question_counter = 0

aptitude_responses_counter = 0 # Load the scaler and model

scaler = joblib.load('1_scaler.pkl') # Loading the scaler
```

```
model = joblib.load('1_svm_model.pkl') # Loading the trained model @app.route('/')

def home():

# Initial welcome message displayed on the web page welcome_message = (

"Chatbot: Hello! I am a career counseling expert with a specialization in the Indian education system. I am here to provide guidance and information to help individuals navigate their academic and professional journeys. How can I assist you today?")

return render_template('index.html') def predict_career(avg_scores):

global total_score

# Use the loaded scaler and model to predict the career category avg_scores = list(avg_scores)

print(avg_scores)  total_score = avg_scores[4]

print(f"total score : {total_score}") avg_scores = avg_scores[:4] print(f"average score : {avg_scores}")

features_scaled = scaler.transform([avg_scores]) print(f"scaled features : {features_scaled}")

model_output = model.predict(features_scaled)[0] print(f"model output : {model_output}")

return model_output, total_score
global in_ai_mode in_ai_mode = False

@app.route('/result', methods=['POST']) def result():

global recommendation_mode global user_responses

global ocean_responses

global ocean_question_counter global aptitude_responses_counter global in_ai_mode

# Handling user query submitted through the form user_input = request.form['user_input']

# Chatbot response handling response = ""

if user_input == "ai":

in_ai_mode = True

return "Welcome to AI! How can I help you today?" elif user_input == "exit":
```

```
in_ai_mode = False

return "Exiting AI mode. Now you can ask about capabilities and career recommendation."

elif in_ai_mode:

# Handle user input related to AI conversation chat_completion =

openai_client.chat.completions.create( messages=[

{"role": "user", "content": user_input},
{"role": "system",

"content": "Your name is Mr. Krish. You are a career counselling expert." "You will

be primarily addressing students of class 8 to 10."

"You have a specilization in Indian education system. Answer questions like that."

}

],

model="gpt-3.5-turbo"

)

return chat_completion.choices[0].message.content else:

if "capabilities" in user_input.lower():

return "Chatbot: You can ask me questions about careers related to: science, technology, engineering, medicine, business, arts, law, education, agriculture, design, psychology."

elif "recommend" in user_input.lower():

recommendation_mode = True # Switch to recommendation mode user_responses =

[] # Reset user responses

ocean_responses = [] # Reset ocean responses ocean_question_counter =

0 # Reset ocean question counter return  initial_questions[0]  #  Display

first  initial  question elif "about" in user_input.lower():

return "Chatbot: Hello! I am a career counseling expert with a specialization in the Indian education system. I am here to provide guidance and information to help individuals navigate their academic
```

```
and professional journeys. How can I assist you today?"

elif any(topic in user_input.lower() for topic in ["science", "technology", "tech", "engineering",
"medicine", "healthcare", "business", "management", "arts", "humanities", "law", "legal",
"education", "teaching", "agriculture", "farming", "design", "creative", "psychology",
"counseling"]):
return get_topic_info(user_input.lower()) elif

recommendation_mode:

if len(user_responses) < len(initial_questions):

user_responses.append(int(user_input)) # Store user responses as integers if

len(user_responses) < len(initial_questions):

return initial_questions[len(user_responses)] # Display next initial question else:

# All initial questions have been answered ocean_question_counter = 0 #

Reset OCEAN question counter return ocean[0] # Display first OCEAN

question

recommendation_mode = False # Turn off recommendation mode for now elif

ocean_question_counter < len(ocean):

# Handle OCEAN questions

ocean_responses.append(int(user_input)) # Store OCEAN responses as integers

ocean_question_counter += 1 # Increment OCEAN question counter

if ocean_question_counter < len(ocean):

return ocean[ocean_question_counter] # Display next OCEAN question else:

# All OCEAN questions have been answered

recommendation_mode = False # Turn off recommendation mode for now

calculate_ocean_averages() # Calculate OCEAN averages

return handle_ocean_questions()
```

```
else:

return "Chatbot : Invalid response, please try again with a new prompt!"
def handle_ocean_questions():

global recommendation_mode global

user_responses

# Calculate average scores for initial questions

avg_scores = calculate_initial_question_averages(user_responses) # Call the

model to predict the career category

model_output = predict_career(avg_scores)

print(model_output) recommendation_mode = False

return redirect('/result_page?model_output=' + model_output[0]) def

calculate_initial_question_averages(responses):

global aptitude_responses global

total_score

introvertedness_scores = sum(responses[0:5])

intuitive_scores = sum(responses[5:10]) feeling_scores =

sum(responses[10:15]) perceptive_scores =

sum(responses[15:20])

aptitude_responses = responses[20:] # Remove the last 10 elements from responses and assign
them to aptitude_responses

# Calculate total score by comparing user responses with correct answers total_score =

sum(

1 for user_ans, correct_ans in zip(aptitude_responses, aptitude_answers) if user_ans
== correct_ans)

avg_introvertedness = introvertedness_scores / 5 avg_intuitive

= intuitive_scores / 5
```

```
avg_feeling = feeling_scores / 5
avg_perceptive = perceptive_scores / 5

# Reset aptitude responses after calculating averages aptitude_responses =

[]

return avg_introvertedness, avg_intuitive, avg_feeling, avg_perceptive, total_score def

calculate_ocean_averages():

global ocean_responses ocean_averages =

[]

for i in range(0, len(ocean_responses)-10, 10):

ocean_subset = ocean_responses[i:i + 10]

ocean_averages.append(sum(ocean_subset) / 10) return

ocean_averages


def get_topic_info(topic):

# Logic to provide information based on user's topic of interest if "science"

in topic:

return "Chatbot: Science-related career paths include biology, chemistry, physics, astronomy, environmental science, and geology."

elif any(subtopic in topic for subtopic in ["technology", "tech"]):

return "Chatbot: Technology careers span software development, data science, cybersecurity, network engineering, artificial intelligence, and robotics."

elif "engineering" in topic:

return "Chatbot: Engineering offers diverse fields like civil, mechanical, electrical, chemical, aerospace, and biomedical engineering."

elif any(subtopic in topic for subtopic in ["medicine", "healthcare"]):

return "Chatbot: Medical professions encompass doctors, nurses, pharmacists, dentists, physical therapists, and healthcare administrators."
```

```
elif any(subtopic in topic for subtopic in ["business", "management"]):
return "Chatbot: Business paths cover marketing, finance, human resources, entrepreneurship, supply chain management, and international business."

elif any(subtopic in topic for subtopic in ["arts", "humanities"]):

return "Chatbot: Arts and humanities fields involve literature, history, languages, visual arts, performing arts, philosophy, and cultural studies."

elif any(subtopic in topic for subtopic in ["law", "legal"]):

return "Chatbot: Legal careers include lawyers, paralegals, judges, legal consultants, and specialists in various areas such as corporate law or criminal law."

elif any(subtopic in topic for subtopic in ["education", "teaching"]):

return "Chatbot: Education paths cover teaching, school administration, educational psychology, curriculum development, and special education."

elif any(subtopic in topic for subtopic in ["agriculture", "farming"]):

return "Chatbot: Agriculture offers careers in farming, agribusiness, agricultural engineering, agronomy, agricultural economics, and sustainable agriculture."

elif any(subtopic in topic for subtopic in ["design", "creative"]):

return "Chatbot: Design and creative fields include graphic design, interior design, fashion design, animation, user experience design, and game design."

elif any(subtopic in topic for subtopic in ["psychology", "counseling"]):  return "Chatbot:

Psychology careers involve clinical psychology, counseling
psychology, industrial-organizational psychology, neuropsychology, and forensic
psychology." else:

return "Chatbot: I'm sorry, I don't have information about that topic." @app.route('/result_page')

def result_page():

global ocean_averages

model_output = request.args.get('model_output', '')



# Calculate OCEAN averages
ocean_averages = calculate_ocean_averages() print(ocean_averages)
```

```
# Create personality labels

personality_labels = ['Openness', 'Conscientiousness', 'Extraversion', 'Agreeableness', 'Neuroticism']

# Create a custom color palette

custom_palette = ['#3498DB', '#2ecc71', '#e74c3c', '#9b59b6', '#f1c40f']  # Create a bar chart with

custom colors and labels

ax = sns.barplot(x=personality_labels, y=ocean_averages, palette=custom_palette)  # Rotate x-axis

labels

ax.set_xticklabels(ax.get_xticklabels(), rotation=25, ha='right')  # Add text annotations to the bars

for i, score in enumerate(ocean_averages):

ax.text(i, score, str(score), ha='center', va='bottom', color='black', fontweight='bold')  # Add a title

and axis labels

plt.title("OCEAN Personality Scores") plt.ylabel("Score")

# Save the plot to a static image file image_path = 'static/ocean_score.png' plt.savefig(image_path,

bbox_inches='tight') # Clear the plot to avoid memory leaks plt.clf()

# Return the path to the static image file
return render_template('result.html', image_path=image_path, model_output=model_output, total_score=model_output[1])

if __name__ == '__main__': app.run(debug=True)
```

CHAPTER 6

TESTING

CHAPTER 6

TESTING

Testing is an integral part of our project development process, ensuring that the chatbot system meets the required standards of functionality, reliability, and usability. In this chapter, we detail the testing methodologies employed in our project, including black box testing, white box testing, unit testing, integration testing, and end-to-end testing.

1. Black Box Testing: Black box testing was conducted to assess the functionality of the chatbot system from an end-user perspective. Test cases were designed based on expected inputs and outputs, and the system's behavior was evaluated against these test cases. By focusing solely on the system's inputs and outputs, black box testing helped identify discrepancies between expected and actual behavior, enabling us to rectify any issues and ensure the system's functionality aligned with user expectations.
2. White Box Testing: White box testing involved analyzing the internal logic and code structure of the chatbot system. Testers had access to the source code and used this knowledge to design test cases that exercised different paths and conditions within the code. This allowed us to verify the correctness of the underlying algorithms and logic implemented in the system, ensuring that it performed as intended under various conditions.
3. Unit Testing: Unit testing was performed to validate individual components or units of code within the chatbot system. Test cases were written to test each function, class, and module in isolation, ensuring that they behaved as expected. By identifying and fixing defects at an early stage of development, unit testing helped improve code quality and maintainability, leading to a more robust and reliable system overall.
4. Integration Testing: Integration testing focused on testing the interactions and interfaces between different components or modules of the chatbot system. This ensured that the integrated units functioned correctly together as a whole. By validating the communication protocols, data exchange mechanisms, and overall system behavior during integration, we were able to identify and address any issues arising from the interaction between subsystems.
5. End-to-End Testing: End-to-end testing encompassed testing the entire chatbot system from start to finish to ensure that it met the specified requirements and behaved as

expected in a real-world environment. Test scenarios were designed to simulate user interactions with the system, assessing its overall performance, reliability, and usability. End-to-end testing helped validate the system's functionality across all its components, ensuring a seamless user experience.

6.1 Black Box Testing

In the project, we conducted black box testing to ensure the functionality of the chatbot system met the specified requirements without looking into its internal implementation. This type of testing focused solely on the external behavior of the application, treating it as a "black box" where the internal workings were unknown.

One of the key test cases, TC-001, involved verifying that the welcome message was displayed correctly upon accessing the home page. This test ensured that users were greeted appropriately when interacting with the chatbot. Upon executing the test, the expected welcome message was indeed displayed, confirming the successful functioning of this essential feature.

Similarly, TC-002 aimed to validate the chatbot's response when asked about its AI capabilities. By inputting a query regarding the system's AI capabilities, we expected the chatbot to provide relevant information. Upon execution, the chatbot appropriately responded with details about its AI capabilities, demonstrating its ability to handle user queries effectively.

TC-003 focused on testing the initiation of the career recommendation mode. Upon receiving a specific user query requesting career recommendations, the chatbot was expected to enter this mode. The test confirmed that the chatbot successfully entered the career recommendation mode, ensuring that users could seamlessly access this feature.

Furthermore, TC-004 examined the chatbot's ability to provide information about specific career paths. By inputting queries related to particular career fields, such as engineering, the chatbot was expected to deliver relevant details. The test outcome indicated that the chatbot effectively retrieved and presented information about engineering careers, enhancing its usability as a career counseling tool.

Lastly, TC-005 assessed the chatbot's response to invalid user input. In scenarios where users input queries that did not match any predefined actions, the chatbot was expected to handle such cases gracefully by displaying an error message. The test confirmed that the chatbot appropriately handled invalid user input, ensuring a smooth user experience even in unforeseen circumstances.

Test Case ID	Test Case Description	Test Input	Expected Output	Actual Output	Pass/Fail
TC-001	Verify welcome message is displayed upon accessing home.	Accessing home page	"Hello! I am a career counseling expert …"	"Hello! I am a career counseling expert …"	Pass
TC-002	Test user query related to AI capabilities.	Input: "What are your AI capabilities?"	Display information about AI capabilities	Displayed information about AI capabilities	Pass
TC-003	Test user query related to career recommendation	Input: "Recommend a career"	Initiate career recommendation mode	Initiated career recommendation mode	Pass
TC-004	Test user query for information about a specific career	Input: "Tell me about engineering careers"	Display information about engineering careers	Displayed information about engineering careers	Pass
TC-005	Test invalid user input	Input: "c"	Display error message	Displayed error message	Pass

Table 6.1.: Black box testing description and results

6.2 White Box Testing

White box testing, also known as structural testing or glass box testing, involves examining the internal structure and logic of the software application. white box testing primarily focused on analyzing the functionality of individual components and their interactions within the system. The objective was to verify that each function, method, or module behaves as expected and handles different inputs and scenarios appropriately.

Functional Testing: Functional testing aimed to validate the behavior of specific functionalities or features within the application. Test cases were designed to verify that each function or feature performs its intended operations accurately and produces the expected outputs.

Functional testing covered various aspects such as user input validation, data processing, output generation, and error handling.

Code Path Analysis: Code path analysis involved examining the different execution paths within the application's codebase. Test cases were designed to ensure that all possible code paths, including conditional branches and loops, were tested thoroughly. By analyzing code paths, testers could identify any potential logic errors, boundary conditions, or edge cases that might lead to unexpected behavior or software failures.

Boundary Value Testing: Boundary value testing focused on testing the boundaries or limits of input variables or parameters within the application. Test cases were designed to evaluate how the system behaves when inputs are at the lower, upper, or boundary values of valid ranges. This testing approach helps uncover issues related to data truncation, overflow, or incorrect handling of boundary conditions.

Test Case ID	Description	Expected Outcome	Actual Outcome	Pass/Fail
TC-001	User Input Validation	Validate that the application rejects invalid inputs and displays appropriate error messages.	All invalid inputs are rejected, and relevant error messages are displayed.	Pass
TC-002	Data Processing	Verify that the application processes user inputs correctly and generates accurate responses.	User inputs are processed accurately, and responses are generated without errors.	Pass
TC-003	Output Generation	Ensure that the application generates the expected output based on the provided inputs.	Correct outputs are generated as per the input scenarios.	Pass
TC-004	Error Handling	Validate that the application handles errors gracefully and provides helpful error messages.	Errors are handled appropriately, and informative error messages are displayed to users.	Pass

Table 6.2.: White box testing description and results

6.3 Unit Testing

Unit testing plays a critical role in ensuring the functionality and reliability of software applications. By testing individual units or components of code in isolation, developers can

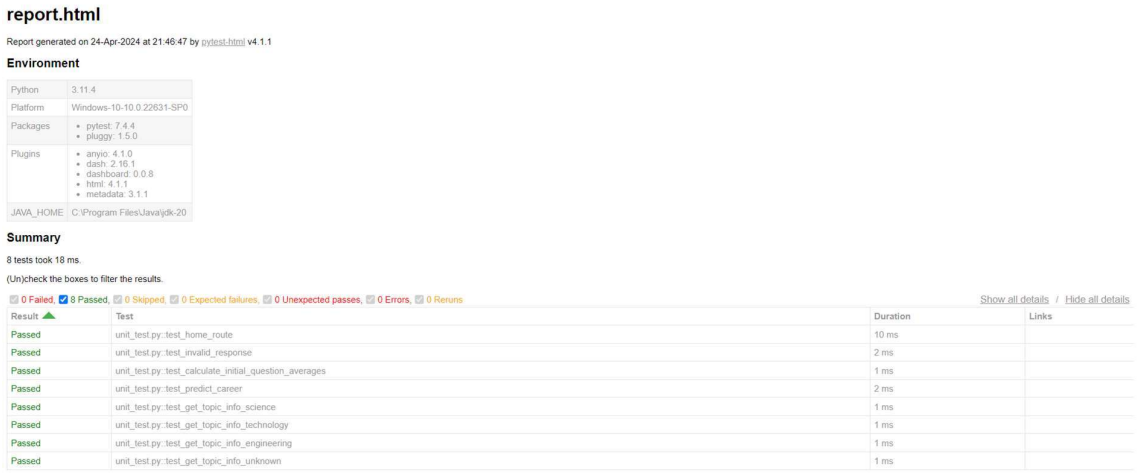
identify and address defects early in the development process. This section presents a description of the unit tests conducted for the project, highlighting the specific functionalities tested and their outcomes.

Test Case ID	Test Name	Description	Pass/Fail
TC-001	test_home_route	This test verifies the functionality of the home route of the application, ensuring that it returns a successful response (HTTP status code 200).	Pass
TC-002	test_invalid_response	This test validates the handling of invalid user responses by the application, ensuring that an appropriate message is displayed when an invalid response is received.	Pass
TC-003	test_calculate_initial_question_averages	This test evaluates the accuracy of the function responsible for calculating initial question averages based on user responses. It checks whether the calculated averages match the expected values.	Pass
TC-004	test_predict_career	This test assesses the prediction functionality of the application, ensuring that it correctly predicts a career based on the provided average scores.	Pass
TC-005	test_get_topic_info_science	This test verifies the retrieval of information related to science careers, ensuring that the application provides accurate and relevant information.	Pass
TC-006	test_get_topic_info_technology	Similar to the previous test, this one evaluates the retrieval of information related to technology careers.	Pass
TC-007	test_get_topic_info_engineering	This test assesses the retrieval of information related to engineering careers.	Pass
TC-008	test_get_topic_info_unknown	This test examines the application's response when a request is made for information on an unknown topic, ensuring that it handles such requests gracefully.	Pass

Table 6.3.: Unit testing description and results

Each test case description outlines the specific functionality being tested, while the corresponding test result indicates whether the test passed successfully. These tests were executed using the Pytest framework, and the results demonstrate the robustness and reliability

of the individual components within the application. Furthermore, the unit testing process ensures that the project maintains high standards of quality and functionality throughout its development lifecycle.



report.html

Report generated on 24-Apr-2024 at 21:46:47 by pytest-html v4.1.1

Environment

Python	3.11.4
Platform	Windows-10-10.0.22631-SP0
Packages	• pytest: 7.4.4 • pluggy: 1.5.0
Plugins	• anyio: 4.1.0 • dash: 2.16.1 • dashboard: 0.0.8 • html: 4.1.1 • metadata: 3.1.1
JAVA_HOME	C:\Program Files\Java\jdk-20

Summary

8 tests took 18 ms.

(Un)check the boxes to filter the results.

0 Failed, 8 Passed, 0 Skipped, 0 Expected failures, 0 Unexpected passes, 0 Errors, 0 Reruns — Show all details / Hide all details

Result	Test	Duration	Links
Passed	unit_test.py::test_home_route	10 ms	
Passed	unit_test.py::test_invalid_response	2 ms	
Passed	unit_test.py::test_calculate_initial_question_averages	1 ms	
Passed	unit_test.py::test_predict_career	2 ms	
Passed	unit_test.py::test_get_topic_info_science	1 ms	
Passed	unit_test.py::test_get_topic_info_technology	1 ms	
Passed	unit_test.py::test_get_topic_info_engineering	1 ms	
Passed	unit_test.py::test_get_topic_info_unknown	1 ms	

Fig.6.1.: Pytest html report for unit testing

6.4 Integration Testing

Integration testing evaluates the interaction between different components or modules of a system to ensure they function correctly together. In the context of this project, integration testing focuses on verifying the seamless integration of various features within the application, such as data loading, response handling, and overall user experience.

Test Case ID	Test Name	Description	Pass/Fail
TC-001	test_data_loading	This test verifies the loading of the application's homepage, ensuring that it returns a successful response (HTTP status code 200) and contains the expected content indicating the presence of the career counseling expert.	Pass
TC-002	test_data_loading_1	Similar to the previous test, this one focuses on the loading of the homepage but specifically checks for the presence of the "Career Chatbot" banner, ensuring its proper display on the page.	Pass
TC-003	test_responcea_handling	This test evaluates the handling of user input by the application when a query related to its capabilities is submitted. It ensures that the application responds appropriately with information about the available career-related questions.	Pass

Table 6.4.: Integration testing description and results

The integration test cases provide insights into the functionality and integration of different components within the application, contributing to a comprehensive assessment of its performance. The results of these tests were generated using pytest HTML, with the report image shown below

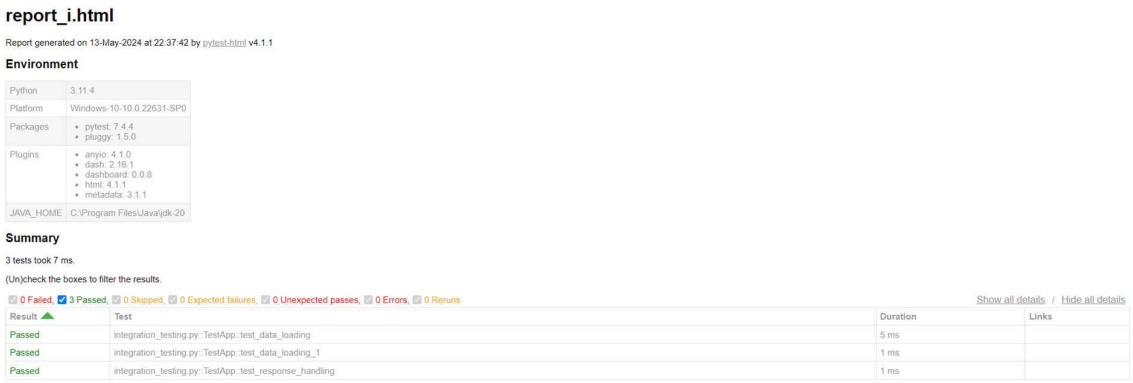


Fig.6.5.: Pytest html report for integration testing

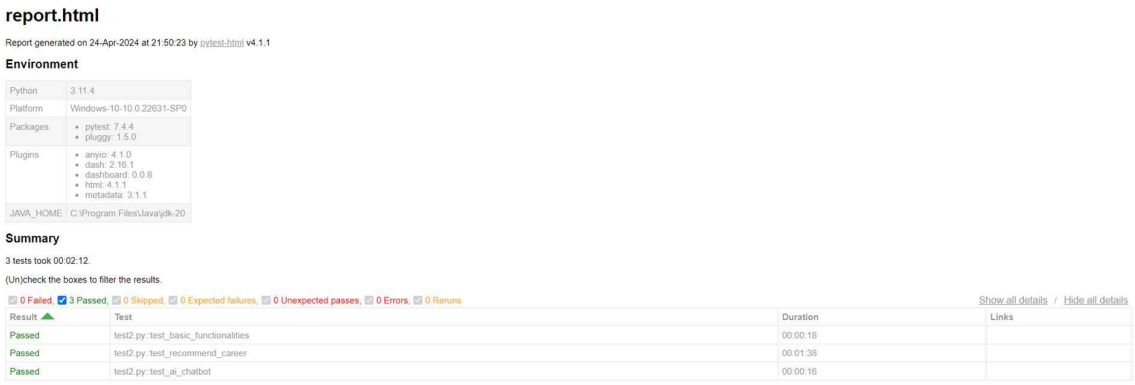
6.5 End to End Testing

End-to-end testing assesses the entire application workflow, simulating real user interactions from start to finish. In this project, end-to-end testing ensures the seamless functionality of core features such as basic functionalities, career recommendation, and interaction with the AI chatbot.

Test Case ID	Test Name	Description	Pass/Fail
TC-001	test_basic_functionalities	This test verifies the basic functionalities of the application, including the loading of the homepage, user interaction with the chatbot, and the display of initial responses.	Pass
TC-002	test_recommend_career	This test evaluates the functionality of the career recommendation feature, ensuring that the chatbot accurately recommends careers based on user responses to a series of questions.	Pass
TC-003	test_ai_chatbot	This test focuses on testing specific commands and questions directed to the AI chatbot, ensuring accurate responses and interaction flow. information about the available career-related questions.	Pass

Table 6.6.: End to End testing description and results

The end-to-end test cases provide insights into the holistic performance of the application, encompassing user interactions and feature functionalities. Additionally, the test results are generated using `pytest-html`, which provides detailed HTML reports showcasing test execution logs and screenshots for reference and analysis. These reports offer a comprehensive overview of test outcomes, aiding in identifying and addressing any issues encountered during testing.



report.html

Report generated on 24-Apr-2024 at 21:50:23 by pytest-html v4.1.1

Environment

Python	3.11.4
Platform	Windows-10-10.0.22631-SP0
Packages	• pytest: 7.4.4 • pluggy: 1.5.0
Plugins	• anyio: 4.1.0 • dash: 2.16.1 • dashboard: 0.0.8 • html: 4.1.1 • metadata: 3.1.1
JAVA_HOME	C:\Program Files\Java\jdk-20

Summary

3 tests took 00:02:12.

(Un)check the boxes to filter the results.

0 Failed, 3 Passed, 0 Skipped, 0 Expected failures, 0 Unexpected passes, 0 Errors, 0 Reruns — Show all details / Hide all details

Result	Test	Duration	Links
Passed	test2.py::test_basic_functionalities	00:00:18	
Passed	test2.py::test_recommend_career	00:01:38	
Passed	test2.py::test_ai_chatbot	00:00:16	

Fig.6.3: Pytest html report for End to End testing

6.6 AI Model Evaluation

Model evaluation is the process of assessing the performance and effectiveness of a machine learning model using various metrics and techniques. It involves comparing the model's predictions to actual outcomes on unseen data to determine its accuracy, precision, recall, F1-score, and other relevant measures. The goal of model evaluation is to understand how well the model generalizes to new data and to identify any issues or areas for improvement before deploying the model in production.

The classification report provides detailed insights into the performance of the trained SVM model on the test dataset. It includes precision, recall, F1-score, and support for each personality type.

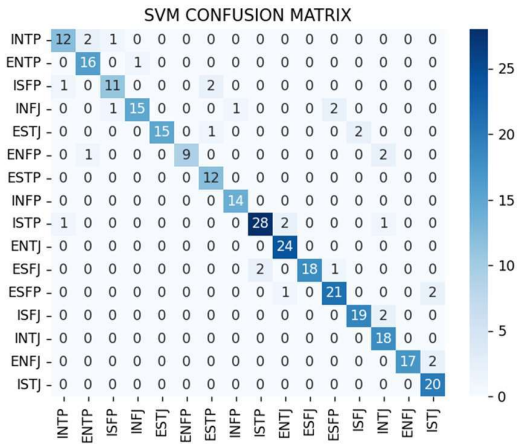


Fig 6.4: SVM confusion matrix

Accuracy: 90%

Macro Average F1-Score: 0.89

Weighted Average F1-Score: 0.90

The overall performance of the SVM model is impressive, with an accuracy of 90%. The macro and weighted average F1-scores indicate that the model performs well across different personality types, achieving a balance between precision and recall.

This classification report serves as a comprehensive summary of the model's performance and can be used to make informed decisions regarding its deployment and further optimization.

CHAPTER 7

EXPERIMENTAL RESULTS

CHAPTER 7

EXPERIMENTAL RESULTS

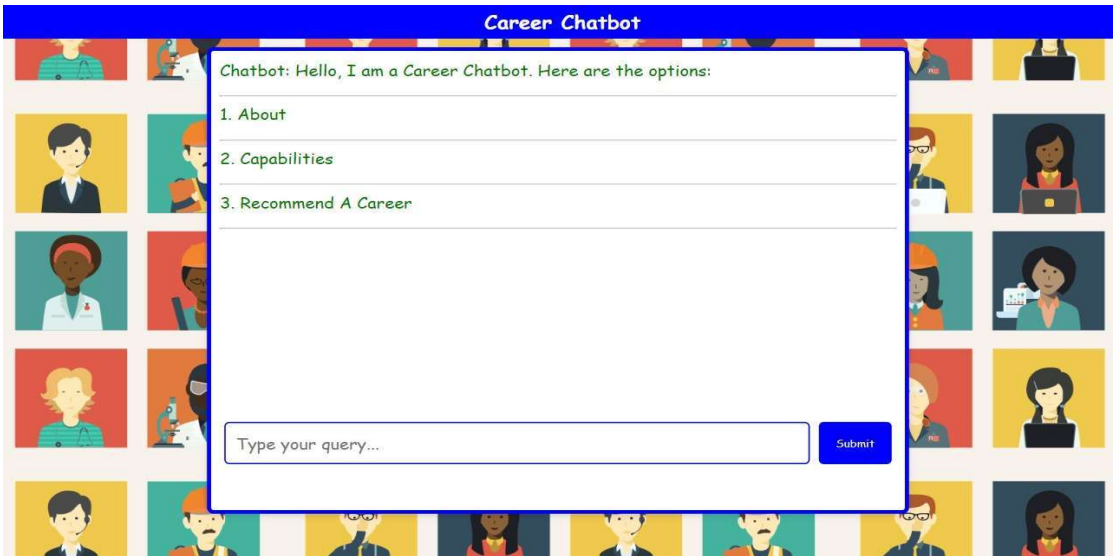


Fig 7.1: Landing page and Initial response from chat bot.

This is the landing page of the Career chatbot. As soon as the user enters the system , the user can see the options ‘About’ , ‘Capabilities’ and ‘Recommend a Career’.

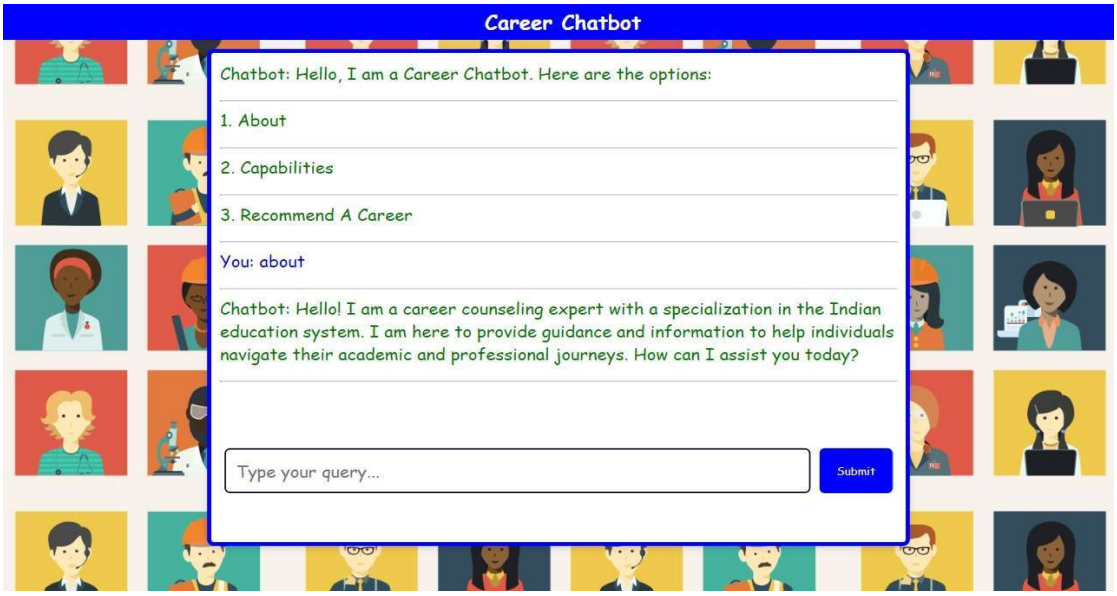


Fig 7.2: Checking basic functionality: About

When the user selects “About” in the query area we can see the details of the chatbot and the purpose of building this system.

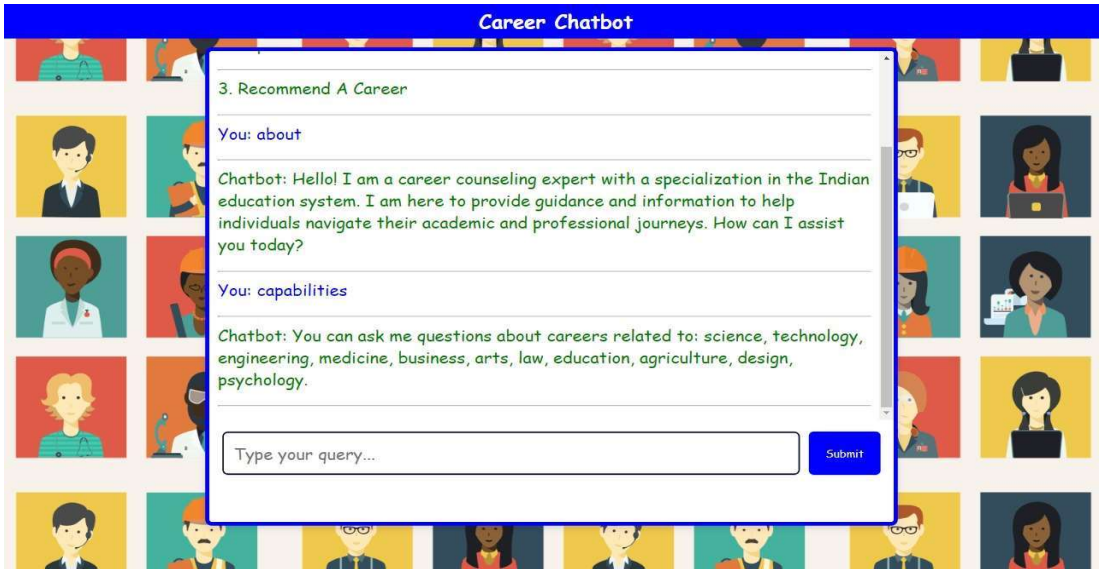


Fig 7.3: Checking basic functionality: Capabilities.

When the user selects “capabilities” in the query area we can see the capabilities of the system and also the fields in which a career can be recommended is also visible.

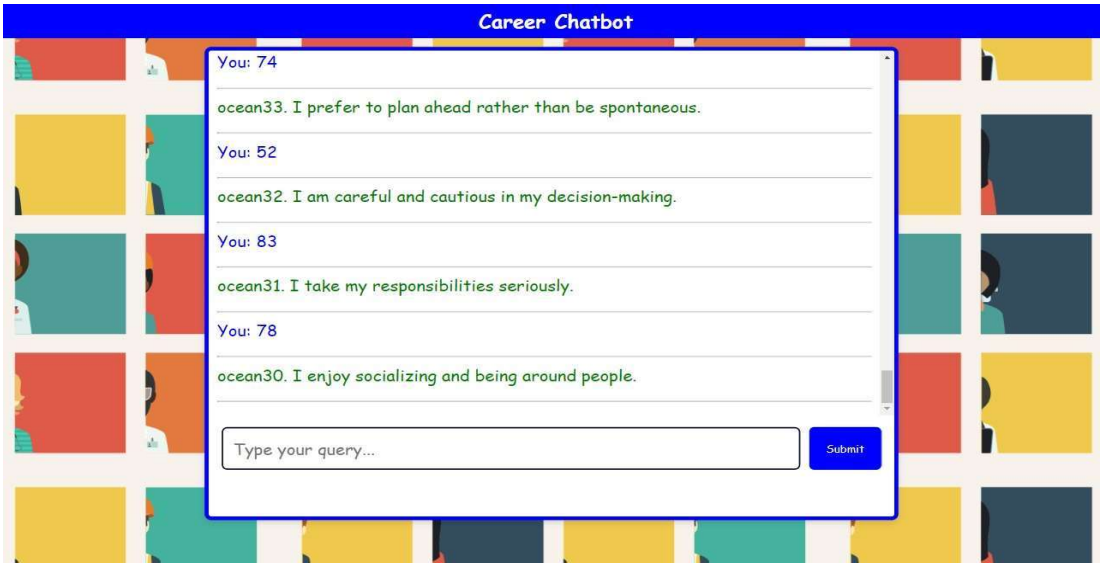


Fig 7.4: Answering questions for career guidance.

When the user press “recommend a career” in the query area , the systems starts asking the questions which are to be answered within 0-100 .There are three types of questions to be answered by the user ‘MBTI’ , ‘OCEAN’ and ‘Aptitude’ questions.

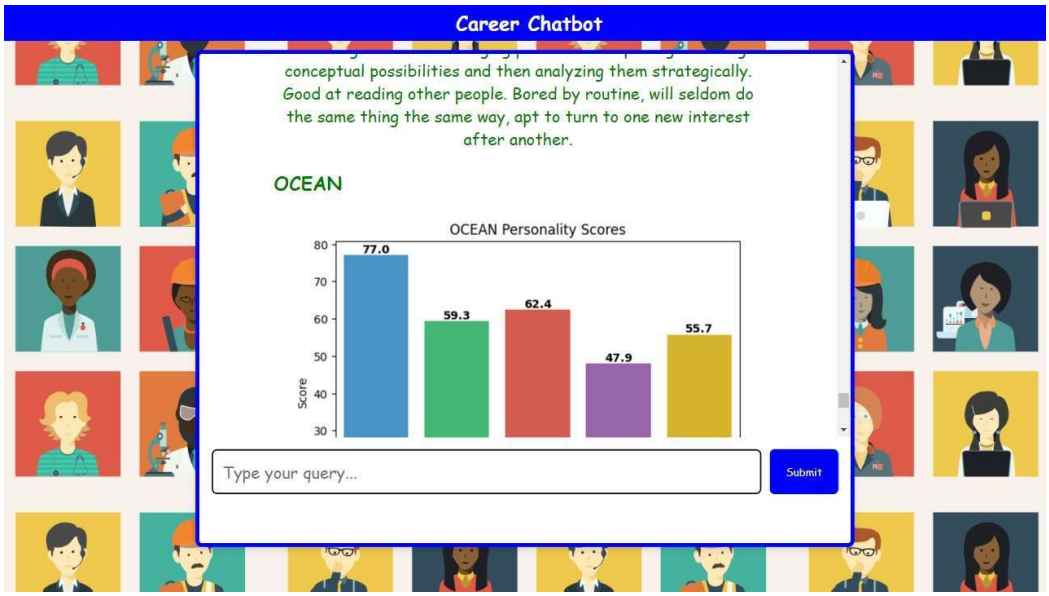


Fig 7.5.: Result of the career guidance test

After the completion of the test , the students are given a personality score and the career suitable for their personality is recommended. For personality detection, we are using MBTI and OCEAN technology.

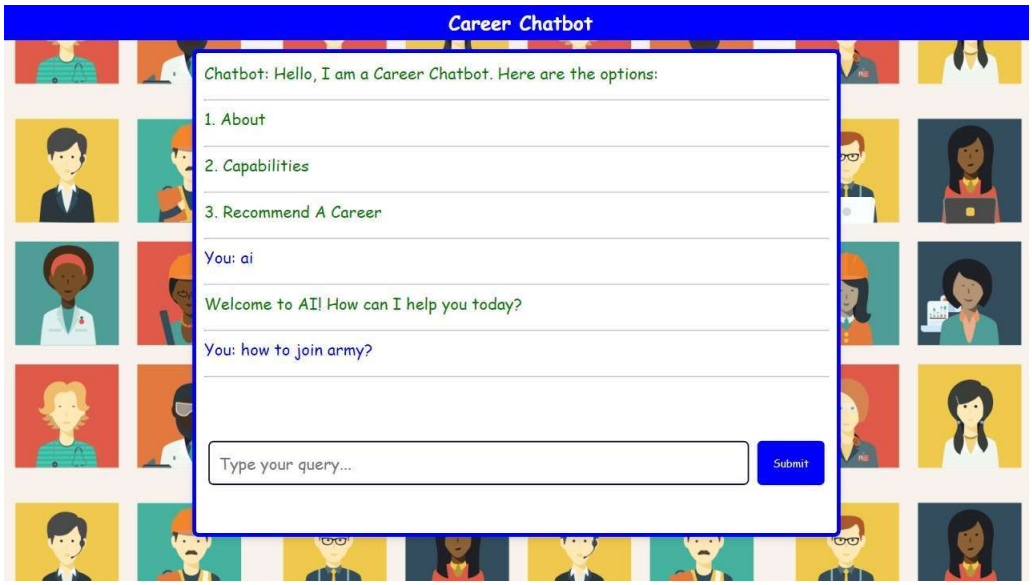


Fig 7.6.: Switching to AI mode

"AI" is to be typed in the chatbot to switch to AI mode . Here the student can ask queries about a particular career based and how to pursue it . It is designed particularly for the Indian Education System.

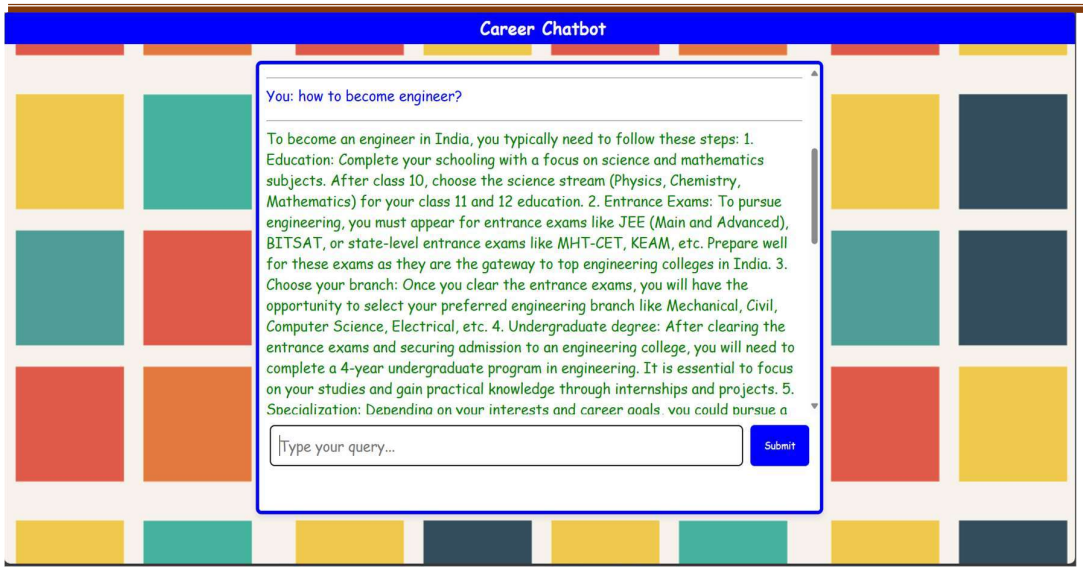


Fig 7.7.: Response to Users queries

The chatbot gives a detailed answer to the question asked by the user ,making it easy for the user to select a career path.

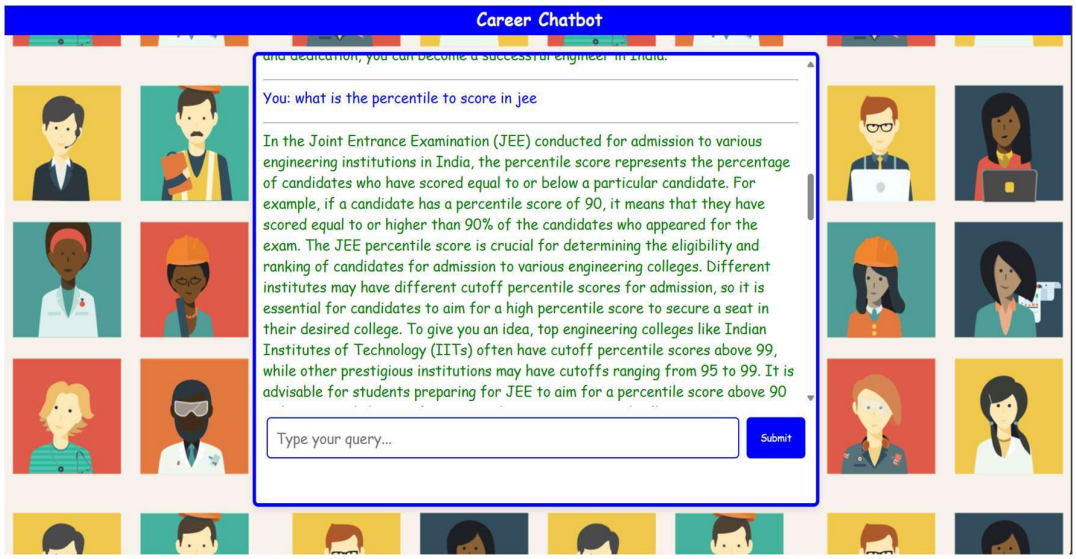


Fig 7.8.: Personalized queries of the student

The user can ask any kind of query regarding the career and path to achieve it.

CHAPTER 8

CONCLUSION

CHAPTER 8

CONCLUSION

The AI-based career counseling project represents a significant step forward in leveraging technology to provide personalized career guidance and support to individuals. Through the development and implementation of an interactive chatbot system, users can access tailored advice, explore career options, and receive relevant information based on their preferences and interests.

Throughout the project lifecycle, various testing methodologies, including unit testing, integration testing, and end-to-end testing, were employed to ensure the robustness and reliability of the application. These testing processes helped identify and rectify potential issues, ensuring a seamless user experience and enhancing the overall quality of the product.

Furthermore, the incorporation of AI algorithms and natural language processing techniques enabled the chatbot to understand user queries effectively and provide accurate responses, thereby enhancing user engagement and satisfaction. By continuously refining and updating the chatbot's knowledge base, the system can adapt to evolving career trends and user preferences, ensuring relevance and usefulness over time.

The AI-based career counseling project holds immense potential in revolutionizing traditional career guidance methodologies, offering a scalable and accessible solution to individuals seeking career advice and direction. Moving forward, continued research and development efforts will be essential in further enhancing the capabilities and effectiveness of the chatbot system, ultimately empowering individuals to make informed decisions about their career paths.

CHAPTER 9

FUTURE ENHANCEMENT

CHAPTER 9

FUTURE ENHANCEMENT

As the landscape of career counseling continues to evolve, there are several avenues for enhancing the AI-based career counseling project to better serve the needs of users and provide more comprehensive support. Below are some potential future enhancements that can further elevate the project's effectiveness and impact:

1. Personalization and Customization:

- Implement advanced machine learning algorithms to analyze user preferences, behaviors, and historical interactions with the chatbot.
- Utilize this data to tailor recommendations and advice more precisely to each user's unique needs and career aspirations.
- Offer personalized learning pathways, career assessments, and skill development opportunities based on individual strengths and weaknesses.

2. Expanded Career Database and Resources:

- Continuously update and expand the chatbot's knowledge base with the latest information on emerging industries, job roles, and career pathways.
- Integrate APIs and web scraping techniques to gather real-time data from reputable sources such as job boards, industry reports, and educational institutions.
- Provide access to a wider range of resources, including articles, webinars, workshops, and online courses, to support users in their career exploration and development journey.

3. Integration with Learning Management Systems (LMS):

- Partner with educational institutions, employers, and training providers to integrate the chatbot with existing learning management systems.
- Enable seamless transfer of user data, such as skills assessments, course completion records, and certifications, between the chatbot and LMS platforms.
- Facilitate personalized learning recommendations and career planning based on users' educational backgrounds, professional goals, and learning preferences.

4.Enhanced Natural Language Understanding (NLU):

- Invest in advanced natural language processing (NLP) techniques to improve the chatbot's ability to understand complex queries, context, and user intent.
- Implement sentiment analysis algorithms to gauge users' emotions and tailor responses accordingly, providing empathetic and supportive interactions.
- Integrate voice recognition capabilities to enable users to interact with the chatbot through voice commands and speech input, enhancing accessibility and user experience.

5. Career Coaching and Mentorship Features:

- Introduce virtual career coaching and mentorship programs, where users can connect with experienced professionals and receive one-on-one guidance and advice.
- Facilitate networking opportunities and peer support groups within the chatbot platform, allowing users to collaborate, share insights, and learn from each other's experiences.
- Incorporate feedback mechanisms to gather user testimonials, reviews, and suggestions for improving the chatbot's services and functionalities.

6. Mobile Applications and Multi-platform Accessibility:

- Develop dedicated mobile applications for iOS and Android devices, providing users with convenient access to the chatbot anytime, anywhere.
- Ensure cross-platform compatibility and synchronization of user data across different devices and operating systems to maintain a seamless user experience.
- Explore integration with popular messaging apps and social media platforms to reach a broader audience and facilitate engagement through familiar communication channels.

7. Ethical and Responsible AI Practices:

- Prioritize ethical considerations such as user privacy, data security, and algorithmic transparency in the development and deployment of AI-based features.
- Implement robust data governance frameworks and compliance measures to protect sensitive user information and ensure regulatory compliance with relevant data protection laws.

- Provide clear disclosures and consent mechanisms regarding data collection, usage, and sharing practices, empowering users to make informed decisions about their privacy preferences.

By embracing these future enhancements, the AI-based career counseling project can evolve into a more sophisticated and user-centric platform, empowering individuals to navigate the complexities of the modern job market and pursue fulfilling and rewarding careers. Through continuous innovation and collaboration with stakeholders, the project can remain at the forefront of digital career guidance solutions, driving positive outcomes for users and contributing to the advancement of the field of career counseling.

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